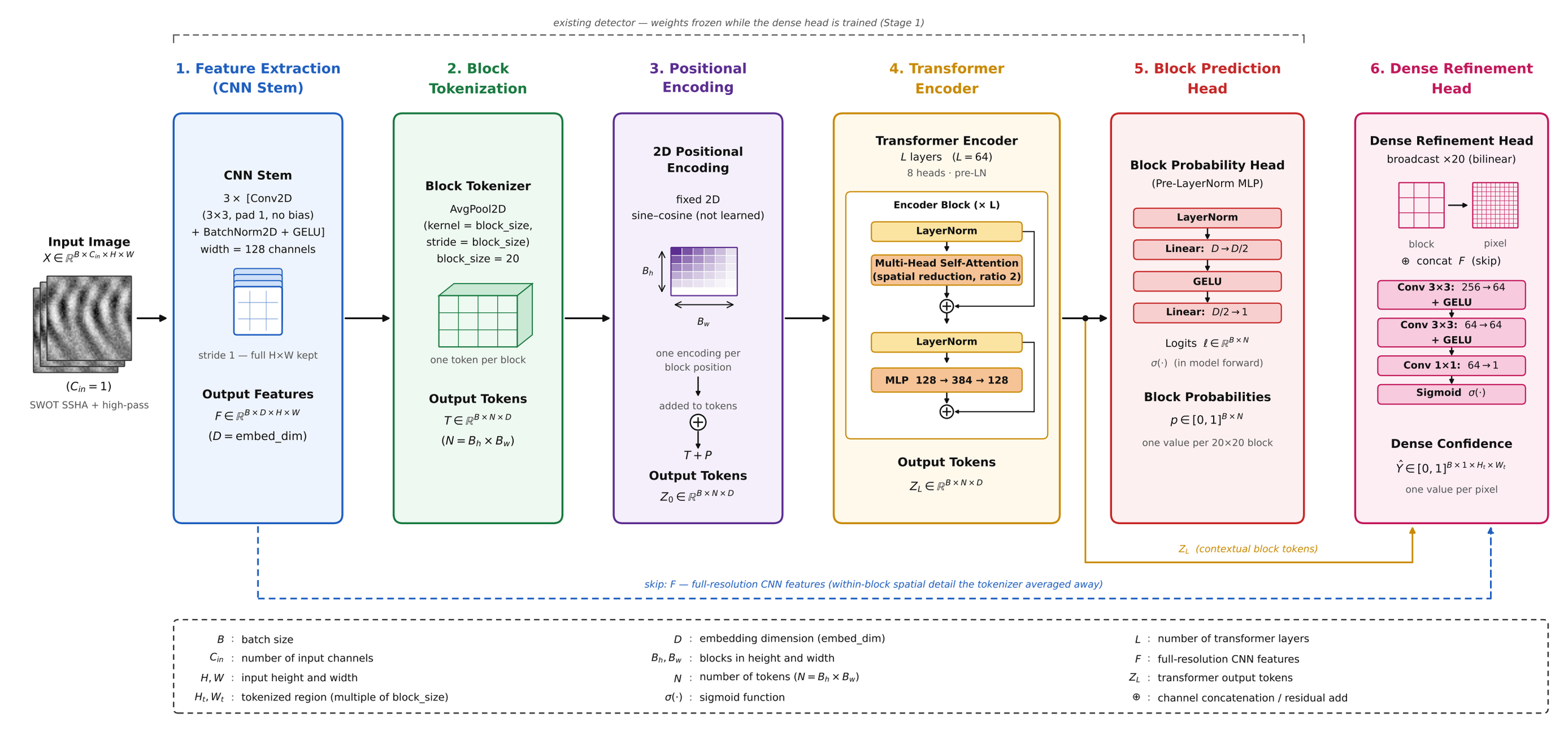

existing detector — weights frozen while the dense head is trained (Stage 1)
1. Feature Extraction (CNN Stem)
2. Block Tokenization
3. Positional Encoding
4. Transformer Encoder
5. Block Prediction Head
6. Dense Refinement Head
Input Image
$X \in \mathbb{R}^{B \times C_{in} \times H \times W}$
$(C_{in} = 1)$
SWOT SSHA + high-pass
CNN Stem
3 × [Conv2D (3×3, pad 1, no bias) + BatchNorm2D + GELU]
width = 128 channels
stride 1 — full H×W kept
Output Features
$F \in \mathbb{R}^{B \times D \times H \times W}$
$(D = \text{embed_dim})$
Block Tokenizer
AvgPool2D (kernel = block_size, stride = block_size)
block_size = 20
one token per block
Output Tokens
$T \in \mathbb{R}^{B \times N \times D}$
$(N = B_h \times B_w)$
2D Positional Encoding
fixed 2D sine–cosine (not learned)
B_h
B_w
one encoding per block position
added to tokens
$T + P$
Output Tokens
$Z_0 \in \mathbb{R}^{B \times N \times D}$
Transformer Encoder
L layers $(L = 64)$
8 heads · pre-LN
Encoder Block (× L)
LayerNorm
Multi-Head Self-Attention (spatial reduction, ratio 2)
LayerNorm
MLP 128 → 384 → 128
Output Tokens
$Z_L \in \mathbb{R}^{B \times N \times D}$
Block Probability Head
(Pre-LayerNorm MLP)
LayerNorm
Linear: $D \to D/2$
GELU
Linear: $D/2 \to 1$
Logits $\ell \in \mathbb{R}^{B \times N}$
$\sigma(\cdot)$ (in model forward)
Block Probabilities
$p \in [0,1]^{B \times N}$
one value per 20×20 block
Dense Refinement Head
broadcast ×20 (bilinear)
block
pixel
⊕ concat F (skip)
Conv 3×3: 256 → 64 + GELU
Conv 3×3: 64 → 64 + GELU
Conv 1×1: 64 → 1
Sigmoid $\sigma(\cdot)$
Dense Confidence
$\hat{Y} \in [0,1]^{B \times 1 \times H_t \times W_t}$
one value per pixel
Z_L (contextual block tokens)
skip: F — full-resolution CNN features (within-block spatial detail the tokenizer averaged away)
B : batch size
C_{in} : number of input channels
H, W : input height and width
H_t, W_t : tokenized region (multiple of block_size)
D : embedding dimension (embed_dim)
B_h, B_w : blocks in height and width
N : number of tokens ($N = B_h \times B_w$)
$\sigma(\cdot)$: sigmoid function
L : number of transformer layers
F : full-resolution CNN features
Z_L : transformer output tokens
⊕ : channel concatenation / residual add